StreamClarify: Proactive Clarification for Grounded Streaming Video Question Answering

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Abstract

Streaming video question answering (QA) requires real-time reasoning over continuous video streams under strict latency and active-context budgets. Current streaming vision-language models (VLMs) often give confident yet ungrounded answers when queries are ambiguous or have false premises, or when relevant visual evidence is no longer in the active context. Proactive clarification can help, but must satisfy the same latency and memory constraints. We present StreamClarify, a training-free framework for grounded streaming QA that combines a pre-answer uncertainty check with hierarchical memory. The check uses the entropy of a binary pre-answer judgment as a lightweight uncertainty score to decide whether to answer directly or request clarification. To support re-answering, a temporal-pyramid KV cache maintains a bounded active visual context, while a CPU-side semantic retrieval archive provides access to historical frames. We further construct StreamClarify-QA, a diagnostic benchmark of 1,837 QA pairs from StreamBench and OVO-Bench, spanning standard, referential ambiguity, and unanswerable scenarios. Across three VLM backbones, StreamClarify improves Macro accuracy by up to 8.2 points over backbone-matched baselines. At comparable trigger rates, entropy-based selection achieves more than 18 points higher precision than random selection in identifying initially incorrect answers. Under oracle user guidance, clarification improves Macro accuracy by up to 3.5 points. We also report a controlled Qwen3-VL simulation on Ambiguous QA with generated questions and referent-only user replies, where the entropy-gated condition improves accuracy by 8.3 points.

1. Introduction

Streaming video question answering (QA) places vision-language models (VLMs) in an always-on assistant setting, requiring real-time reasoning over continuous, unsegmented video streams [4, 36]. Unlike offline video QA on pre-segmented clips [10, 30, 33], streaming assistants process frames in a single online pass within a bounded active-context budget. Recent advances in memory-efficient long-horizon inference [7, 16, 21, 43] and proactive interaction [4, 24, 28, 40] have significantly advanced this frontier. However, most existing frameworks implicitly assume that (i) the user query is well-specified and (ii) the visual evidence needed to answer is available in the current context.

In practice, both assumptions can fail. Under a bounded active-context budget, visual content relevant to a later query may be compressed or removed from the active context. Users may also pose imperfect queries, including referential ambiguity (*e.g.*, "where did I leave it?") and false premises (*e.g.*, asking about an object that never appeared), which can prevent reliable grounding even when related frames remain visible. Although these failures have different causes, current models tend to answer by default, producing fluent but ungrounded responses, a common form of hallucination [2, 20, 31]. A useful streaming assistant should instead decide whether to answer immediately or ask for clarification. Clarification can identify an ambiguous referent or correct a false premise.

Making this work in streaming environments raises two challenges. First, deciding whether to answer directly or request clarification must be cheap: established uncertainty estimates often require multiple samples or repeated forward passes, which exceed a per-frame latency budget, while trained ambiguity detectors may not transfer across backbones or domains [9, 14, 17]. Second, a clarified query may refer to video content that is no longer represented in the bounded active context. A practical system therefore needs a low-overhead uncertainty check and access to earlier video content for re-answering.

To address these requirements, we present *StreamClarify*, a training-free framework that augments a frozen streaming VLM with two components. First, a pre-answer uncertainty check is applied before answer generation. The model makes a binary `[Yes]/[No]` judgment about whether its answer would rely on observed video evidence

rather than general knowledge alone, and StreamClarify uses the entropy of this judgment as a lightweight uncertainty signal. If the entropy exceeds a fixed threshold, answer generation is suspended and clarification is triggered. Second, hierarchical memory supports re-answering after clarification. The system keeps a small active window of recent frames at full detail, compresses older content to stay within a fixed context budget, and maintains a searchable CPU-side archive of historical frames. Once clarified user guidance is available, StreamClarify uses it to search the archive, prefills the retrieved frames alongside the active context, and re-answers.

Our contributions are:

- We propose *StreamClarify*, a training-free proactive clarification framework for frozen streaming VLMs. It combines a pre-answer uncertainty check with hierarchical memory to suspend answer generation, request clarification, and retrieve historical frames before re-answering.
- We introduce *StreamClarify-QA*, a diagnostic streaming benchmark of 1,837 QA pairs spanning standard, referential-ambiguity, and unanswerable scenarios, designed to expose answer-by-default behavior under imperfect queries in causal streaming conditions.
- Experiments across three streaming VLM backbones show Macro accuracy gains of up to 8.2 points over backbone-matched baselines while operating in real time. Additional experiments analyze the pre-answer uncertainty check, clarification effectiveness, and active-context management.

2. Related Work

2.1. Streaming Video Understanding

Unlike offline video understanding that processes pre-segmented clips in batches [10, 30, 33], streaming architectures process continuous frames in a single online pass. These approaches broadly follow two directions. The first focuses on memory-efficient inference within a bounded memory: recent works manage the active context through attention sinks [36], compress representations via token merging or summarization [5, 16, 26, 35, 43], or offload historical representations to CPU memory for on-demand retrieval [6, 7, 15, 21, 23, 45], drawing on KV cache techniques originally developed for text-only LLMs [3, 34, 38]. The second direction investigates proactive interaction: models like VideoLLM-Online [4], LiveStar [40], and StreamMind [8] autonomously determine response timing, while other frameworks adapt offline VLMs via plug-and-play modules [28], asynchronous pipelines [24], or anticipatory planning [39] to enable continuous dialogue, accompanied by emerging online benchmarks [19, 22].

Despite these architectural advances, current streaming systems generally assume well-formed queries with fully observable visual evidence. Neither the memory-efficient nor the proactive interaction line of work addresses model behavior under imperfect queries, a dimension our work directly targets.

2.2. Proactive Clarification and Uncertainty Estimation

Proactive clarification has been explored in dialogue and static visual QA via uncertainty-guided timing [27], future-turn anticipation [44], and supervised policies for resolving visual ambiguity [13], but all of these require task-specific training that limits their applicability to new domains or backbone models. A related line of work examines whether VLMs can correctly avoid answering when questions are unanswerable or based on false premises [12, 32], and hallucination surveys have documented the prevalence of over-confident responses in VLMs [2, 20, 31]. Together, these works highlight that deciding when to suspend answering is a significant open challenge for current VLMs.

For uncertainty estimation, training-free methods including semantic entropy [9, 17] and verbalized self-assessment [14] are proposed to avoid task-specific supervision but rely on multi-sample generation or repeated forward passes. These approaches have not been systematically studied in the streaming setting, where per-frame latency budgets and backbone calibration variability impose additional constraints that remain unaddressed.

3. Methodology

3.1. Problem Setup

We model streaming video QA as a sequence of streaming interactions $\langle V_i\rangle\langle Q_i\rangle\langle A_i\rangle$ $(i = 1, 2, \ldots)$ over a continuous stream $\mathcal{V} = \{F_1, F_2, \ldots\}$, where $\langle V_i\rangle = \mathcal{V}_{\leq t_i}$ denotes the video frames observed up to query time t_i, $\langle Q_i\rangle$ is the user query, and $\langle A_i\rangle$ is the model response at round i. At query time t_i, the model has access only to a memory-constrained representation of $\langle V_i\rangle$ under budget $\mathcal{M}$, so historical frames may be compressed or evicted before a query arrives. In practice, $\mathcal{M}$ is implemented as a token budget $L_{\max}$ for the active KV cache of the streaming backbone.

Beyond memory constraints, users may also issue *imperfect queries*, such as queries with referential ambiguity or false premises. Standard streaming VLMs lack an explicit mechanism to recognize when such queries cannot be grounded. In this work, the core framework uses clarified user guidance and does not depend on a particular question generator. When clarification is requested, the model suspends $\langle A_i\rangle$ and obtains clarified user-side guidance $\langle R_i\rangle$. The model uses $\langle R_i\rangle$ to guide archive retrieval before re-answering. Thus, the interaction is extended from $\cdots\langle V_i\rangle\langle Q_i\rangle\langle A_i\rangle\cdots$ to $\cdots\langle V_i\rangle\langle Q_i\rangle\langle R_i\rangle\langle A_i\rangle\cdots$.

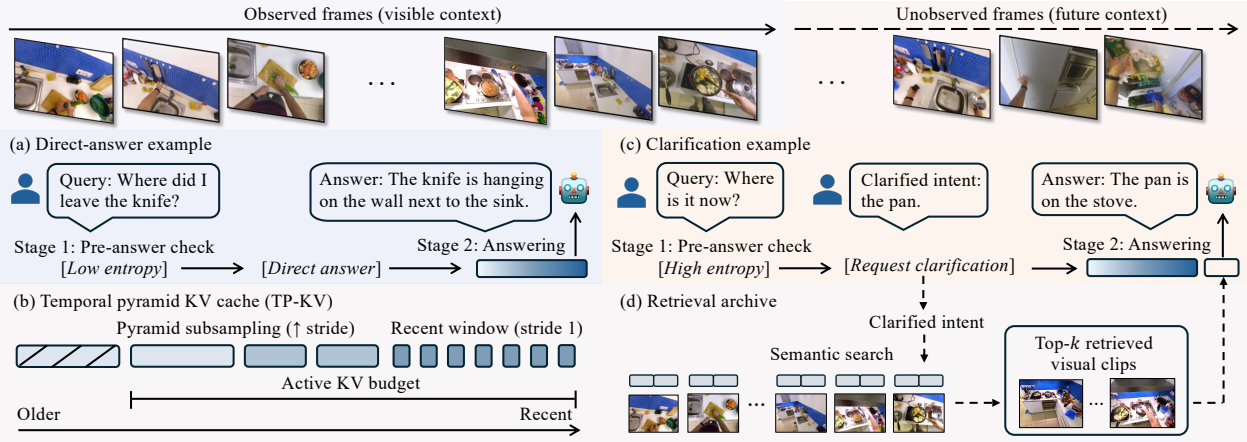


Figure 1. Overview of StreamClarify. The framework augments a frozen streaming VLM with a pre-answer uncertainty check in a two-stage generation loop and hierarchical memory. (a) When pre-answer entropy is below the trigger threshold, the system answers directly using the active temporal-pyramid KV (TP-KV) cache. (b) TP-KV keeps recent frames at full resolution and compresses older observed frames within a fixed active budget. (c) When entropy exceeds the threshold, the system requests clarification and converts the resulting user guidance into a retrieval query. (d) Semantic search retrieves relevant past clips from the archive and combines them with the active TP-KV context for re-answering.

3.2. StreamClarify

As shown in Fig. 1, StreamClarify augments a frozen streaming VLM with two training-free components: a pre-answer uncertainty check for deciding whether to answer directly or request clarification, and a hierarchical memory that maintains a bounded active context while providing access to historical frames.

3.2.1. Pre-Answer Uncertainty Check

StreamClarify implements the pre-answer uncertainty check as a two-stage generation pipeline that shares the same KV cache. Stage 1 asks whether the answer would rely on observed video content rather than general knowledge and emits a binary `[Yes]/[No]` judgment. Stage 2 generates the answer conditioned on that judgment.

We measure uncertainty from the resulting two-label distribution. Let p denote the probability assigned to `[Yes]` at the token position where the backbone distinguishes the labels. The binary entropy is $H = -p\log_2(p) - (1-p)\log_2(1-p)$. If H exceeds a fixed per-backbone threshold set from the no-clarification distribution, answer generation is suspended and clarification is triggered. This requires only one short generation before the answer and no additional training. Supplementary Sec. G lists the prompt templates.

3.2.2. Hierarchical Memory

Hierarchical memory combines a bounded temporal-pyramid KV cache (TP-KV) with a CPU-side retrieval archive. Direct answers use only TP-KV. After clarification, the user guidance queries the archive for frames no longer retained in the active context.

TP-KV for bounded active context. TP-KV keeps recent frames at full granularity and compresses older history with a temporal pyramid. Each incoming frame is first converted into backbone vision tokens and prefilled through the model to update the active KV cache. When the cache exceeds the token budget $L_{\max}$, TP-KV protects a recent window, compresses older tokens by stride sampling at progressively coarser pyramid levels, and evicts the oldest tokens only after the compressed history still exceeds the allocated budget controlled by the compression ratio ρ. The retained tokens are then ordered chronologically and prefilled to rebuild a bounded cache. This preserves recent detail while compactly representing earlier frames.

Retrieval archive for historical frames. For each incoming frame, the archive stores backbone vision tokens for later prefilling and a compact image-text key for semantic search. After clarification, a search-query prompt maps the user guidance into the same retrieval space. The top-K historical clips are then prefilled through the backbone and combined with the current TP-KV context before re-answering. Supplementary Algorithm 1 in Sec. B details the online TP-KV update and archive construction procedure.

The archive grows with streamed time and is measured separately from the bounded active context. For longer streams, a retention window can bound archive storage without changing the TP-KV budget. Supplementary Sec. F analyzes scalability on an approximately 30-minute stream.

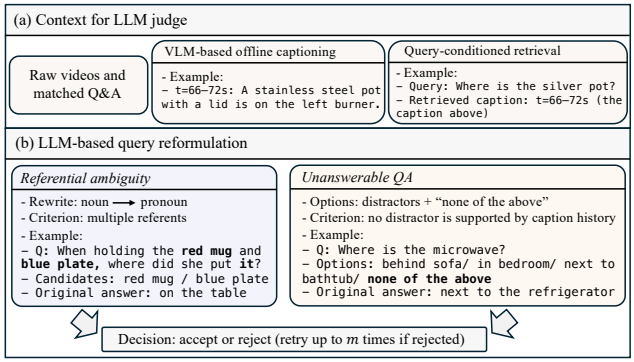


Figure 2. StreamClarify-QA construction pipeline. (a) Dense captions are aligned to each query timestamp. (b) Eligible QA pairs are transformed into Referential Ambiguity or Unanswerable variants, followed by type-specific validation. OVO-Bench HLD questions are retained unchanged.

4. StreamClarify-QA Construction

To systematically evaluate model behavior under imperfect queries, we construct *StreamClarify-QA*, a diagnostic streaming benchmark covering three scenario types. *Standard QA* contains well-specified answerable queries. *Referential Ambiguity QA* covers queries with underspecified references. *Unanswerable QA* requires the model to select an abstention response because the observed video prefix supports none of the provided substantive answer options. The construction pipeline is illustrated in Fig. 2.

4.1. Data Sources

This diagnostic benchmark is built on two complementary datasets, summarized in Table 1(a). *StreamBench* contributes 1,556 deduplicated QA pairs from 260 videos (avg. 253 s) in open-ended format. *OVO-Bench* provides questions in multiple-choice format. We select all videos containing at least one Hallucination Detection (HLD) question, together with all other questions from those videos. OVO-Bench defines HLD as questions irrelevant to the available video input [22]. To better reflect realistic streaming interaction patterns where queries are naturally spaced over time, questions with a temporal gap of less than 5 seconds from the preceding question are removed, retaining 281 QA pairs from 95 videos (avg. 383 s). All questions are evaluated causally at their annotated timestamps, so the model can use only the video prefix available when each question arrives.

4.2. Query Transformation

For each source video, MiniCPM-V 4.5 [42] generates dense captions at 2 FPS and 448-pixel resolution from overlapping 6-second windows with a 3-second stride. The caption prompt includes the source questions but not their answers, and the resulting captions are aligned to each query timestamp. Questions answerable from general world knowledge are excluded, leaving 1,302 eligible pairs from StreamBench and 130 non-HLD pairs from OVO-Bench. Unanswerable construction additionally excludes 307 yes/no StreamBench questions, leaving 995 source pairs. All transformations are performed using Qwen3-14B [37].

Referential Ambiguity. An LLM rewrites each eligible query by replacing a specific noun phrase with an ambiguous pronoun, subject to the criterion that multiple plausible referents exist in the caption context at query time. Invalid rewrites are retried up to 3 times. This yields 351 and 53 candidate pairs from StreamBench and OVO-Bench respectively, before human quality filtering. Pairs that exhaust their retry budget are preserved as Standard QA.

Unanswerable QA. Inspired by the native HLD category in OVO-Bench [22], we construct a complementary Unanswerable split for StreamBench and combine the two sources in StreamClarify-QA. Each remaining open-ended StreamBench question is converted into multiple-choice format by generating plausible distractors. Each candidate set is checked against the caption history up to the query timestamp. If any distractor is supported by the captions, the set is rejected and regenerated for up to three attempts. After validation, the source answer is removed and "None of the above" is added. The 151 OVO-Bench HLD questions are retained unchanged, while the 389 constructed StreamBench instances undergo caption-based filtering and raw-video label validation in Sec. 4.3. Together, they form 540 Unanswerable QA pairs.

Transformations are applied independently to individual QA pairs while preserving their original timestamps in the continuous video stream. OVO-Bench does not annotate linked dialogue turns, while StreamBench transformations do not deliberately select or link earlier QA turns that identify the intended referent. We therefore evaluate clarification of the current query without treating prior dialogue as a defined benchmark input.

4.3. Quality Control and Dataset Statistics

Timestamp-aligned captions guide both query transformations, followed by query-specific quality checks. Because ambiguity depends on the final wording, human review of all 404 Referential Ambiguity candidates removes insufficiently ambiguous rewrites and cases whose referent remains recoverable from the original annotations, leaving 302 pairs. For StreamBench Unanswerable QA, caption-based filtering and balanced split assembly retain 389 instances. To account for visual information that captions may omit, we screen all 389 against the causal video prefix available at the query timestamp. Three rounds of Gemini [11] screening leave seven cases for human review,

Table 1. Statistics of StreamClarify-QA. (a) Video and QA overview by source. (b) Scenario distribution across both datasets. Abbreviations: Vid = videos, Dur. = duration, Ref. Ambig. = referential ambiguity, and Unans. = unanswerable.

(a) Overview

Dataset	#Vid	#QA	Q/Vid	Dur.(s)
StreamBench	260	1,556	6.0	∼253
OVO-Bench	95	281	∼3.0	∼383

(b) Scenario distribution

Scenario	Total	Stream.	OVO
A. Standard	995	913	82
B. Ref. Ambig.	302	254	48
C. Unans. QA	540	389	151

which confirms the original label in every case. The 151 OVO-Bench HLD questions are retained unchanged and are not part of this audit. Supplementary Sec. D.1 reports the filtering counts and raw-video protocol.

The final diagnostic benchmark comprises 1,837 QA pairs across 355 videos: 995 Standard, 302 Referential Ambiguity, and 540 Unanswerable QA instances, as detailed in Tab. 1(b). Imperfect-query scenarios account for nearly half of all instances, placing substantial pressure on robustness beyond standard QA.

5. Experiments

5.1. Settings

Baselines. Baselines include closed-source, training-based, and backbone-matched training-free systems. The closed-source reference is GPT-5-mini [25] evaluated in an offline pre-segmented setting. Training-based baselines include VideoLLM-Online [4], Flash-VStream [43], ReKV [7], and TimeChat [41]. Training-free baselines include StreamChat [35], StreamBridge [28], and StreamingVLM [36], instantiated on the same three backbones as StreamClarify.

Evaluation Metrics. We report Micro Accuracy over all QA, Macro Accuracy averaged across scenario types, time-to-first-token (TTFT), and online processing throughput (ProcFPS). For trigger analyses, trigger rate, Precision, and Recall treat an incorrect initial answer as a positive case, while Error AUROC measures the same ranking across thresholds. Clarification effects are reported through wrong-to-correct and correct-to-wrong transitions. The main, matched-6K, and entropy-gated Generated Clarification comparisons include 95% confidence intervals from paired resampling of complete videos. Correctness is judged by Qwen3-14B [37] using dense offline captions as scene-level reference context.

Implementation Details. StreamClarify and the backbone-matched training-free baselines use three frozen VLMs: LLaVA-OV-1.5 [18], InternVL-3.5 [29], and Qwen3-VL [1], without task-specific training. The main comparison follows each method's standard configuration. StreamClarify uses a 10K-token TP-KV budget, compression ratio 0.25, and a 20 s recency window. We also compare all Qwen3-VL methods under a common 6K active-context budget, excluding CPU-side storage. Clarification uses a per-backbone threshold fixed from the no-clarification setting to obtain a near-30% rate over the pooled 1,837-QA evaluation and retrieves the top-K=16 clips. The evaluation population, backbone, and clarification setting for each analysis are summarized in Supplementary Table 8.

5.2. Main Results

Overall Performance. Table 2 shows that StreamClarify obtains the highest Macro accuracy in all six dataset–backbone settings and the highest Micro accuracy in five. On StreamBench, the Qwen3-VL gains reach 2.3 Micro and 3.3 Macro points. Under InternVL-3.5, Micro accuracy is 1.0 point below StreamBridge, while Macro accuracy is 1.1 points higher. On OVO-Bench, the largest gains are 7.5 points in Micro under Qwen3-VL and 8.2 points in Macro under LLaVA-OV-1.5. Paired video-level intervals exclude zero for the Qwen3-VL Macro gain on StreamBench ($+3.3$ points, 95% CI $[+1.0, +5.7]$) and for the LLaVA-OV-1.5 and Qwen3-VL Macro gains on OVO-Bench ($+8.2$ points, $[+2.2, +13.8]$, and $+7.8$ points, $[+2.4, +12.9]$). Supplementary Sec. D.2 reports the remaining intervals. Against training-based TimeChat, Qwen3-VL StreamClarify gains 12.1 Micro and 15.3 Macro points on StreamBench.

The per-category results in Table 2 show that the clearest gains occur on Unanswerable QA. On OVO-Bench, StreamClarify reaches 68.2%, 56.3%, and 62.9% across the three backbones, improving over their strongest matched baselines by 1.3, 6.6, and 4.6 points. Standard and Ambiguous QA vary more across methods, motivating the clarification analysis in Sec. 5.3. The offline GPT-5-mini reference also drops from 73.4% on Standard QA to 16.9% on Ambiguous and 8.0% on Unanswerable QA on StreamBench, indicating that general model capability alone does not resolve imperfect queries.

System Efficiency. Table 2 reports efficiency in the same no-clarification setting. With Qwen3-VL on StreamBench, StreamClarify processes 10.2 FPS with 0.239 s TTFT, compared with 3.5 FPS and 0.305 s for StreamChat. StreamBridge processes 16.7 FPS but has 2.657 s TTFT, while StreamingVLM has 0.062 s TTFT and retains only its attention sinks and recent window in the active visual context. Clarification-enabled latency is evaluated separately in Sec. 5.3.

5.3. Effectiveness of Proactive Clarification

We evaluate proactive clarification at two levels. The oracle-guided analysis evaluates the entropy trigger together with archive retrieval and re-answering across three question types and three backbones. We then report a generated-question proof of concept on Ambiguous queries with simulated referent-only replies.

Table 2. Per-category streaming video QA results on StreamBench and OVO-Bench in the no-clarification setting using each method's standard configuration. Shaded columns denote the overall accuracy and system-efficiency columns. Greyed text reports the supplementary per-category breakdown (Std. = Standard, Amb. = Ambiguous, UQ = Unanswerable QA). Arrows indicate the preferred direction. Accuracy values are percentages, and TTFT is reported in seconds.

		StreamBench							OVO-Bench						
		Per-Cat. Acc.			**Overall Acc.**		**System**		Per-Cat. Acc.			**Overall Acc.**		**System**	
		Std.	Amb.	UQ	**Micro**	**Macro**	TTFT↓	ProcFPS↑	Std.	Amb.	UQ	**Micro**	**Macro**	TTFT↓	ProcFPS↑
Closed	GPT-5-mini	73.4	16.9	8.0	47.8	32.8	–	–	58.5	54.2	25.2	39.9	46.0	3.829	–
Open	VideoLLM-Online	49.5	11.8	0.5	31.1	20.6	0.064	9.43	13.4	22.9	0.7	8.2	12.3	0.066	11.44
	Flash-VStream	62.1	11.4	17.2	42.6	30.2	0.149	1.97	45.1	47.9	16.6	30.2	36.5	0.125	1.91
	ReKV	67.1	16.5	6.2	43.6	29.9	0.181	4.53	58.5	47.9	30.5	41.6	45.6	0.053	6.76
	TimeChat	64.4	14.2	29.3	47.4	36.0	2.045	5.80	45.1	37.5	37.1	39.5	39.9	1.481	7.60
LLaVA-OV	StreamChat	64.0	15.4	46.5	51.7	41.9	0.700	2.28	53.7	35.4	66.9	57.7	52.0	0.642	2.45
	StreamBridge	73.1	21.7	19.0	51.2	37.9	1.673	4.80	68.3	43.8	44.4	51.2	52.1	2.137	7.46
	StreamingVLM	70.0	22.0	26.2	51.2	39.4	0.092	1.79	61.0	47.9	43.0	49.1	50.6	0.113	1.80
	Ours	64.5	16.5	45.2	**51.9**	**42.1**	0.313	4.98	64.6	47.9	68.2	**63.7**	**60.3**	0.334	5.57
InternVL	StreamChat	63.5	14.6	40.1	49.7	39.4	0.594	3.25	52.4	39.6	36.4	41.6	42.8	0.514	4.06
	StreamBridge	74.3	19.3	39.1	**56.5**	44.2	2.894	9.43	57.3	41.7	49.7	50.5	49.6	3.056	26.95
	StreamingVLM	71.1	16.1	26.2	50.9	37.8	0.072	3.08	52.4	43.8	19.9	33.5	38.7	0.078	3.20
	Ours	67.6	14.6	53.7	55.5	**45.3**	0.240	5.53	59.8	50.0	56.3	**56.2**	**55.3**	0.258	5.89
Qwen3-VL	StreamChat	60.0	16.1	62.7	53.5	46.3	0.305	3.50	45.1	33.3	58.3	50.2	45.6	0.191	3.79
	StreamBridge	75.7	23.6	35.7	57.2	45.0	2.657	16.68	56.1	43.8	37.7	44.1	45.9	2.857	36.93
	StreamingVLM	68.3	21.7	54.0	57.1	48.0	0.062	8.70	59.8	33.3	43.0	46.3	45.4	0.069	9.33
	Ours	67.4	19.3	67.4	**59.5**	**51.3**	0.239	10.16	58.5	39.6	62.9	**57.7**	**53.7**	0.260	12.38

Oracle-Guided Analysis. Table 3 evaluates three backbones with oracle guidance, providing a controlled upper bound on user-guidance quality. For Referential Ambiguity QA, it supplies the correct referent. For Standard and transformed Unanswerable QA, it derives a user-side pointer from the original pre-transformation instance. Naturally unanswerable queries use the question itself as the retrieval query. These inputs are used only after a trigger and do not change the initial active KV cache or insert the reference answer into the answering context. This analysis tests whether the trigger ranks initial-answer errors and whether the system can use correct guidance when it is available.

Table 3 shows that entropy-based triggering yields higher Micro and Macro Accuracy than both no clarification and matched-rate random triggering for all three backbones under oracle guidance. With Qwen3-VL, Micro accuracy is 61.3%, compared with 59.2% without clarification and 60.4% with random triggering. At comparable trigger rates near 33%, entropy obtains 59.8% Precision and 48.5% Recall for initial-answer errors, compared with 41.6% and 34.4% for random. These accuracy gains come with additional query-time cost. For Qwen3-VL, mean TTFT increases from 0.242 to 1.019 s at the 33.0% trigger rate. Section 5.4 isolates the overhead of the pre-answer check, and Supplementary Sec. F breaks down the latency of generated clarification.

Generated-Question Proof of Concept. We further evaluate generated clarification on the 302 Ambiguous QA pairs from StreamBench and OVO-Bench using Qwen3-VL and simulated referent-only replies. We reuse the numerical threshold from the oracle-guided analysis without retuning. It triggers on about one-third of the pooled 1,837-QA evaluation and on 162/302 (53.6%) Ambiguous queries. Direct Retrieval, Generated Clarification, and Oracle Guidance start from the same reconstructed causal state and use the same retrieval and re-answering procedure. *w/o Clar.* retains the frozen original online answer. *Direct Retrieval*

Table 3. Oracle-guided analysis on StreamBench and OVO-Bench pooled (1,837 QA) at comparable trigger rates near 30%. Guidance is provided only after a trigger. Grey text shows per-category accuracy, while shaded columns report overall accuracy and efficiency. Trig., Prec., and Rec. use initially incorrect answers as positives. TTFT and ProcFPS are averaged over all queries.

	Overall Performance							Clarification		
Method	Std.	Amb.	UQ	**Micro**	**Macro**	TTFT↓	ProcFPS↑	Trig.	Prec.	Rec.
LLaVA-OV-1.5										
w/o Clar.	64.5	21.5	51.7	53.7	45.9	0.316	5.14	0.0	–	–
w/ Random	65.6	24.8	49.3	54.1	46.6	1.145	4.05	27.8	44.5	26.7
w/ Entropy	66.9	31.1	49.8	**56.0**	**49.3**	1.166	4.04	28.4	**63.0**	**38.5**
InternVL-3.5										
w/o Clar.	66.9	20.2	54.4	55.6	47.2	0.243	5.63	0.0	–	–
w/ Random	68.2	24.8	55.0	57.2	49.4	0.979	4.33	31.3	43.7	30.8
w/ Entropy	68.4	26.5	57.2	**58.2**	**50.7**	0.976	4.31	30.9	**56.7**	**39.5**
Qwen3-VL										
w/o Clar.	66.6	22.5	66.1	59.2	51.8	0.242	10.75	0.0	–	–
w/ Random	67.0	28.1	66.1	60.4	53.8	1.030	7.57	33.8	41.6	34.4
w/ Entropy	66.8	30.8	68.1	**61.3**	**55.3**	1.019	7.51	33.0	**59.8**	**48.5**

uses the ambiguous query without user guidance, *Generated Clarification* asks one short question, and *Oracle Guidance* directly receives the intended referent.

In the all-query Generated Clarification condition, question generation is attempted without entropy selection. The generator receives the ambiguous query and causal visual state, but not the intended referent, original unambiguous question, reference answer, or future frames. An evaluation-only GPT-5 gate [25] then checks each generated question against the intended referent. It accepts 212 questions and routes the remaining 90 to Direct Retrieval. The same fallback rule is applied within the entropy-gated condition. On accepted items, Generated Clarification and Oracle Guidance receive the same fixed referent-only reply, `I mean {referent}.`, and follow the same downstream retrieval and re-answering path. Under entropy gating, this generation-and-gate procedure is applied to the 162 selected queries, while the original answer is retained for the remaining 140. The all-query conditions bypass entropy selection. Exact prompts and gate details are provided in Supplementary Sec. G.2.

As shown in Table 4, entropy-gated Generated Clarification improves accuracy from 22.5% to 30.8% (+8.3 points, 95% CI $[+4.2, +12.5]$), with 34 corrections and 9 harms. Direct Retrieval yields only one additional correct answer, showing that retrieval and re-answering alone do not explain the gain. Without entropy selection, Generated Clarification reaches 34.8%, compared with the 38.4% upper bound under correct-referent guidance. The gated gain remains positive under separate blinded GPT-5 and Gemini 3.6 Flash answer judges (Supplementary Sec. D.3). Detailed clarification latency is reported in Supplementary Sec. F.

5.4. Analysis and Ablations

We analyze pre-answer scoring, TP-KV, and clarification parameters. The pre-answer scoring and TP-KV ablations use Qwen3-VL on OVO-Bench without clarification or retrieval. The retrieval ablation uses the same dataset and backbone under oracle user guidance. The matched-6K comparison instead evaluates Qwen3-VL on the full StreamBench and OVO-Bench with clarification disabled.

Table 4. Generated-question analysis on 302 Ambiguous QA pairs with Qwen3-VL. Selection specifies entropy-gated or all-query application. W→C and C→W denote paired correctness transitions.

		Overall Accuracy		Paired Change		
Condition	Selection	Correct	Micro (%)	W→C	C→W	ΔCorrect
w/o Clar.	None	68	22.5	–	–	–
Generated Clar.	Entropy-gated	**93**	**30.8**	**34**	**9**	**+25**
Direct Retrieval	All-query	69	22.8	20	19	+1
Generated Clar.	All-query	105	34.8	55	18	+37
Oracle Guidance	All-query	116	38.4	62	14	+48

Table 5. Comparison of pre-answer scoring strategies on OVO-Bench with Qwen3-VL without clarification or retrieval.

Method	Macro (%)	TTFT (s)
Direct Answer	49.3	0.085
Probability	50.3	0.365
V/K Markers	51.3	0.210
Ours (Single)	51.0	0.100
Ours (Two-stage)	**53.7**	0.260

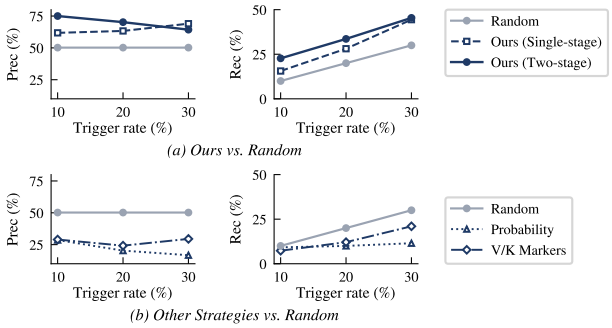


Figure 3. Trigger Precision and Recall for Stage-1 formulations at controlled trigger rates on OVO-Bench with Qwen3-VL. Initial-answer correctness defines the error label.

Pre-Answer Scoring Strategy. Table 5 and Figure 3 compare our two-stage design with four alternatives. Direct Answer prompts the model to answer immediately, without a separate pre-answer judgment. Ours (Single) produces the binary judgment and answer in one generation, while Probability requests an explicit confidence score and V/K Markers substitutes arbitrary tokens for the semantic decision labels. Clarification, retrieval, and re-answering are disabled. Table 5 reports the accuracy and latency of the resulting initial answers, while Figure 3 evaluates how well each score identifies errors in those answers.

Our two-stage design reaches 53.7% Macro Accuracy on its initial answers, 2.4 points above the strongest alternative (Table 5). Across trigger rates, it maintains higher Recall, while Precision remains above random and mostly comparable to the single-stage variant (Figure 3a). The additional pre-answer pass increases TTFT from 0.085 to 0.260 s. Probability and V/K Markers fall below random triggering in Figure 3b, supporting the use of semantically meaningful decision labels instead of arbitrary markers or an explicitly generated probability. The binary-judgment entropy score is also robust. Entropy ranks direct-answer errors better than the directional $p(\mathrm{No})$ baseline on StreamBench, with AUROC of 0.674 versus 0.527. Across changes to prompt wording, demonstration order, and demonstration content, Error AUROC changes by -0.011 to $+0.017$, while score rankings remain strongly correlated with the original prompt (Spearman 0.941–0.978). Full operating-point results are reported in Supplementary Sec. E.

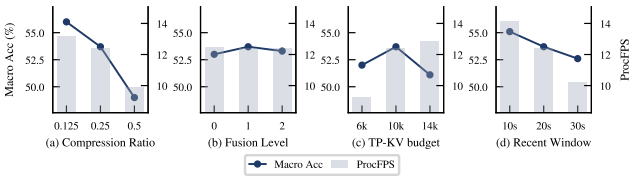


Figure 4. TP-KV hyperparameter ablations on OVO-Bench with Qwen3-VL in the no-clarification setting. Bars show ProcFPS and lines show Macro Accuracy.

Table 6. Matched-6K Qwen3-VL results without clarification. Accuracy is reported in percent, TTFT in seconds, and ProcFPS as online processing throughput.

	StreamBench				OVO-Bench			
Method	Micro	Macro	TTFT↓	ProcFPS↑	Micro	Macro	TTFT↓	ProcFPS↑
StreamChat	52.8	45.4	0.314	3.42	50.5	46.2	0.198	3.68
StreamBridge	57.7	45.7	1.170	24.51	47.7	48.8	1.197	44.86
StreamingVLM	56.7	47.8	0.064	8.86	47.0	46.3	0.071	9.59
Ours	**58.0**	**51.3**	0.232	8.46	**56.9**	**52.0**	0.252	9.23

TP-KV Memory Hyperparameters. We ablate four hyperparameters across two aspects: how tokens are compressed (compression ratio and fusion level), and how the budget is allocated (budget size and recency window size). The results in Figure 4 show that a lower compression ratio improves both Macro Accuracy and ProcFPS, while performance varies only slightly across fusion levels. The budget ablation favors the middle 10K setting, and shorter recency windows allocate more capacity to compressed history while improving Macro Accuracy.

Matched-6K Comparison. For the cross-method comparison, we run all Qwen3-VL methods with a strict 6K active-context ceiling in the no-clarification setting on the full StreamBench and OVO-Bench evaluations. Each method retains its existing memory design. At 2 FPS, the budget represents about 31 s of dense visual history, compared with average stream durations of 253 s and 383 s. For StreamBridge, StreamingVLM, and StreamClarify, compression or eviction occurs in every video, and 84.8–86.7% of queries arrive after the first memory operation. Under strict 6K, StreamClarify compresses 6.19 times per streamed minute and 85.2% of queries arrive after compression, compared with 2.14 and 76.7% under the 10K configuration. The matched-6K setting therefore evaluates most queries after active-context management has already occurred. StreamChat does not maintain a persistent visual KV cache, so the budget is applied to its query-time visual input while its native CPU archive remains unchanged.

Table 6 shows that StreamClarify reports the highest Micro and Macro Accuracy on both benchmarks. Its margins over the strongest baseline on StreamBench are $+0.3$ Micro points (95% CI $[-2.3, +3.0]$) and $+3.4$ Macro points (95% CI $[+1.2, +5.7]$). On OVO-Bench, the corresponding margins are $+6.4$ Micro points (95% CI $[+1.1, +11.8]$) and $+3.2$ Macro points (95% CI $[-3.4, +9.2]$). StreamClarify also maintains real-time processing under this budget, reaching 8.46 FPS with 0.232 s TTFT on StreamBench and 9.23 FPS with 0.252 s TTFT on OVO-Bench.

Clarification Retrieval Parameters. Figure 5 studies retrieval depth and clarification rate under oracle user guidance. Using the same 133/281 triggered OVO-Bench queries, increasing K from 8 to 32 raises Macro Accuracy from 57.2% to 58.9%, while mean TTFT increases from 0.918 to 2.253 s. We retain the intermediate $K{=}16$ setting. Higher clarification rates similarly improve Macro Accuracy while reducing ProcFPS.

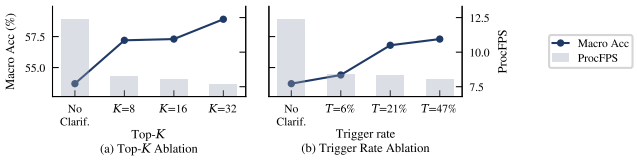


Figure 5. Effect of (a) retrieval depth and (b) clarification rate on OVO-Bench with Qwen3-VL under oracle user guidance. Bars show ProcFPS and lines show Macro Accuracy.

6. Conclusion and Discussion

StreamClarify studies streaming video QA under bounded active context and imperfect queries. Ambiguity can make the intended target uncertain, while false premises can leave the requested information unsupported. StreamClarify combines a training-free uncertainty-based clarification trigger with TP-KV bounded context and clarification-guided archive retrieval. This allows the system to withhold an immediate answer on selected queries and re-answer after receiving focused user guidance.

Several limitations remain. First, the entropy score ranks direct-answer errors but does not identify whether the underlying cause is ambiguity, missing context, or reasoning. Second, generated clarification is evaluated in a single-query setting with a simulated referent-only response. Naturally linked dialogue, noisy responses, and multi-turn interaction are not covered. Third, the current benchmarks do not isolate verified old-evidence recovery, so the memory experiments establish bounded active-context operation rather than a long-video accuracy advantage. Benchmark construction and answer evaluation also use LLM-based judgments, which may introduce bias in borderline cases.

Despite these limitations, StreamClarify takes a step toward streaming assistants that selectively seek focused input before answering. Future work will study natural multi-turn clarification and evaluate memory recovery on long streams with temporally verified supporting evidence.

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